UGANDA CHRISTIAN UNIVERSITY

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G-RAFT: A CLOSED-LOOP GRAPH–RETRIEVAL TUTOR

Grounded, verifiable and adaptive primary mathematics support

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BACKGROUND, RESEARCH GAP & OBJECTIVES

The challenge in AI-supported mathematics

Primary mathematics requires correct steps, curriculum alignment and timely feedback. One-shot LLM answers can sound convincing while hiding reasoning or factual errors.

Hallucinated answers
Confident output may still be mathematically wrong.

Weak curriculum grounding
Generic responses may ignore grade and prerequisite order.

No performance feedback loop
Errors are rarely verified, corrected and reused for improvement.

Why this matters in Uganda

Large classes and limited access to individual tutoring create demand for scalable support. The system aligns with SDG 4, Uganda Vision 2040 and national STEM priorities.

Research aim

Design and evaluate a closed-loop, graph-grounded mathematics tutor that improves answer reliability and adapts from verified feedback.

Objectives

1. Represent curriculum concepts and prerequisites in Neo4j.
2. Retrieve grade-aware evidence from Qdrant.
3. Verify arithmetic and refine weak answers in a loop.
4. Evaluate accuracy, grounding, latency and usability.

METHODOLOGY: G-RAFT CLOSED LOOP

The prototype combines structured curriculum knowledge, semantic retrieval, local generation and deterministic verification. The next phase closes the loop using evaluator and learner feedback.

1 — GROUND — Curriculum Knowledge Graph

Neo4j stores 37 nodes and 39 relationships: grades, topics, concepts, examples and prerequisite paths.

2 — RETRIEVE — Grade-Aware Evidence

Qdrant retrieves relevant learning materials using semantic similarity, grade filters and a confidence threshold.

3 — GENERATE + VERIFY — Hybrid Tutor

Llama 3.2:3B explains grounded context; a restricted arithmetic engine bypasses the LLM for explicit calculations.

4 — FEEDBACK + REFINE — Loop Engineering NEXT PHASE

Check correctness, grounding and learner response. Retry with targeted feedback; log weak cases to improve prompts, retrieval and content.

↻ ASK → GROUND → RETRIEVE → GENERATE → VERIFY → REFINE → LEARN

LITERATURE, PROTOTYPE STATUS & EVALUATION

Key related work

Lewis et al. (2020) — RAG
Combines parametric generation with retrieved external evidence.

Zhang et al. (2024) — RAFT
Trains models to use relevant evidence and ignore distractor documents.

Edge et al. (2024) — GraphRAG
Uses graph structure to support broader, connected retrieval and synthesis.

Madaan et al. (2023) — Self-Refine
Iterative feedback and refinement can improve first-pass LLM outputs.

Working prototype — August 2026

37	39	10	22
Graph nodes	Graph relations	Learning materials	Automated tests

Evaluation plan

Correctness • grounding fidelity • hallucination rate • retrieval relevance
response latency • teacher/learner usability • loop improvement per retry

STATUS: Working grounded tutor; closed-loop refinement is the proposed contribution.